Supplemental Material for "Lifshitz–Kosevich Theory of Anomalous Landau Levels in Topological Flat Bands"

Chao-Xing Liu[1,2]

[1] *Department of Physics, The Pennsylvania State University, University Park, Pennsylvania 16802, USA*

[2] *Center for Theory of Emergent Quantum Matter, The Pennsylvania State University, University Park, Pennsylvania 16802, USA*

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Appendix A: Model and Quantum Geometry Details

A.1. Flat-band condition and the ideal point

We consider the eigen-equation

$$H_\alpha(\mathbf{k})|u_\pm^\alpha(\mathbf{k})\rangle = E_\pm^\alpha(\mathbf{k})|u_\pm^\alpha(\mathbf{k})\rangle, \tag{A.1}$$

and work with the shifted Hamiltonian

$$\widetilde{H}_\alpha(\mathbf{k}) = H_\alpha(\mathbf{k}) - E_0\,\mathbb{I} = \begin{pmatrix} ak^2 & \alpha c k_\alpha \\ \alpha c k_{-\alpha} & \Delta_f \end{pmatrix}, \qquad \Delta_f \equiv E_1 - E_0, \tag{A.2}$$

where $k_\pm = k_x \pm \mathrm{i}k_y$, $k^2 = k_x^2 + k_y^2$, and the two spin sectors $\alpha = \uparrow, \downarrow$ (or equivalently $\alpha = \pm 1$) are related by time-reversal. Starting from Eq. (A.2), the characteristic polynomial is

$$\begin{aligned} \det[\widetilde{H}_\alpha(\mathbf{k}) - \lambda\mathbb{I}] &= (ak^2 - \lambda)(\Delta_f - \lambda) - c^2 k_+ k_- \\ &= \lambda^2 - (ak^2 + \Delta_f)\lambda + (a\Delta_f - c^2)k^2. \end{aligned} \tag{A.3}$$

For a flat band, $\lambda \equiv 0$ for all $\mathbf{k}$. Substituting $\lambda = 0$ gives $(a\Delta_f - c^2)k^2 = 0$, which requires $\Delta_f = c^2/a$, i.e. $E_1 = E_0 + c^2/a$. With this condition, Eq. (A.3) factorises as $\lambda[\lambda - (ak^2 + c^2/a)] = 0$, yielding the band energies

$$E_-^\alpha(\mathbf{k}) = E_0, \qquad E_+^\alpha(\mathbf{k}) = E_0 + ak^2 + \frac{c^2}{a}. \tag{A.4}$$

Bloch states. Define $N(\mathbf{k}) \equiv a^2k^2 + c^2$, and the eigen-wavefunctions are given by

$$|u_-^\alpha(\mathbf{k})\rangle = \frac{1}{\sqrt{N(\mathbf{k})}}\begin{pmatrix} -c \\ \alpha a k_{-\alpha} \end{pmatrix}, \qquad |u_+^\alpha(\mathbf{k})\rangle = \frac{1}{\sqrt{N(\mathbf{k})}}\begin{pmatrix} \alpha a k_\alpha \\ c \end{pmatrix}, \tag{A.5}$$

One verifies the orthonormal condition, e.g. $\langle u_\pm^\alpha | u_\pm^\alpha\rangle = 1$ and $\langle u_+^\alpha | u_-^\alpha\rangle = 0$.

A.2. Quantum geometric tensor

For the isolated flat band in sector α, the projector onto the complementary band is $1 - |u_-^\alpha\rangle\langle u_-^\alpha| = |u_+^\alpha\rangle\langle u_+^\alpha|$. Hence, one can define the quantum geometry tensor as

$$Q_{ij}^\alpha(\mathbf{k}) = \langle \partial_i u_-^\alpha | u_+^\alpha\rangle\langle u_+^\alpha | \partial_j u_-^\alpha\rangle. \tag{A.6}$$

The matrix elements are computed via the Hellmann–Feynman (HF) relation. Define the shifted interband energy

$$\Delta(k) \equiv E_+^\alpha(\mathbf{k}) - E_0 = ak^2 + \frac{c^2}{a}. \tag{A.7}$$

Since $\widetilde{H}_\alpha(\mathbf{k})|u_-^\alpha\rangle = 0$, differentiating with respect to k_i gives $(\partial_i \widetilde{H}_\alpha)|u_-^\alpha\rangle + \widetilde{H}_\alpha|\partial_i u_-^\alpha\rangle = 0$. Taking $\langle u_+^\alpha|$ from the left and using $\langle u_+^\alpha|\widetilde{H}_\alpha = \langle u_+^\alpha|(E_+^\alpha - E_0) = \Delta(k)\langle u_+^\alpha|$ yields

$$\langle u_+^\alpha|\partial_i \widetilde{H}_\alpha|u_-^\alpha\rangle = -\Delta(k)\langle u_+^\alpha|\partial_i u_-^\alpha\rangle. \tag{A.8}$$

The required derivatives of $\widetilde{H}$ are

$$\partial_{k_x}\widetilde{H}_\alpha = \begin{pmatrix} 2ak_x & \alpha c \\ \alpha c & 0 \end{pmatrix}, \quad \partial_{k_y}\widetilde{H}_\alpha = \begin{pmatrix} 2ak_y & ic \\ -ic & 0 \end{pmatrix}. \tag{A.9}$$

Substituting the explicit states Eq. (A.5) (and using $k_{-\alpha}^* = k_\alpha$):

$$(\partial_{k_x}\widetilde{H}_\alpha)|u_-^\alpha\rangle = \frac{1}{\sqrt{N}}\begin{pmatrix} -2ack_x + cak_{-\alpha} \\ -\alpha c^2 \end{pmatrix} = \frac{1}{\sqrt{N}}\begin{pmatrix} -cak_\alpha \\ -\alpha c^2 \end{pmatrix} = -\alpha c|u_+^\alpha\rangle, \tag{A.10}$$

and similarly $(\partial_{k_y}\widetilde{H}_\alpha)|u_-^\alpha\rangle = +\mathrm{i}c\,|u_+^\alpha\rangle$, so that

$$\langle u_+^\alpha|\partial_{k_x}\widetilde{H}_\alpha|u_-^\alpha\rangle = -\alpha c, \qquad \langle u_+^\alpha|\partial_{k_y}\widetilde{H}_\alpha|u_-^\alpha\rangle = +\mathrm{i}c. \tag{A.11}$$

Inserting into Eq. (A.8) gives

$$\langle u_+^\alpha|\partial_{k_x} u_-^\alpha\rangle = \alpha\frac{c}{\Delta(k)}, \qquad \langle u_+^\alpha|\partial_{k_y} u_-^\alpha\rangle = -i\frac{c}{\Delta(k)}. \tag{A.12}$$

(The step $-2ack_x + cak_{-\alpha} = ca(k_x - \mathrm{i}\alpha k_y - 2k_x) = -ca(k_x + \mathrm{i}\alpha k_y) = -cak_\alpha$ in Eq. (A.10).) Inserting into Eq. (A.6) yields the components

$$Q_{xx}^\alpha = \frac{c^2}{\Delta(k)^2}, \quad Q_{yy}^\alpha = \frac{c^2}{\Delta(k)^2}, \quad Q_{xy}^\alpha = -i\alpha\frac{c^2}{\Delta(k)^2}. \tag{A.13}$$

Therefore the quantum metric $g_{ij}^\alpha = \mathrm{Re}\,Q_{ij}^\alpha$ and Berry curvature $\Omega^\alpha = -2\,\mathrm{Im}\,Q_{xy}^\alpha$ are

$$g_{xx}^\alpha = g_{yy}^\alpha = \frac{c^2}{\Delta(k)^2}, \quad g_{xy}^\alpha = 0, \quad \Omega^\alpha = \alpha\frac{2c^2}{\Delta(k)^2}, \tag{A.14}$$

and thus we find the ideal quantum geometry condition

$$\mathrm{tr}\, g^\alpha = \frac{2c^2}{\Delta(k)^2} = |\Omega^\alpha| \tag{A.15}$$

for either spin sector. Substituting $\Delta(k) = ak^2 + c^2/a$ reproduces Eq. (3) of the main text.

A.3. Integrated quantum metric and quantum distance

We consider the momentum cut-off Λ and integration the Berry curvature over the disk $|\mathbf{k}| \le \Lambda$ gives

$$T = \frac{1}{2\pi}\int_0^\Lambda \frac{2c^2}{(ak^2 + c^2/a)^2}\, k\,\mathrm{d}k. \tag{A.16}$$

Substituting $\nu = ak^2 + c^2/a$ (so $\mathrm{d}\nu = 2ak\,\mathrm{d}k$, i.e. $k\,\mathrm{d}k = \mathrm{d}\nu/(2a)$):

$$\begin{aligned} T &= \frac{c^2}{2\pi a}\int_{c^2/a}^{a\Lambda^2 + c^2/a} \nu^{-2}\,\mathrm{d}\nu = \frac{c^2}{2\pi a}\left[-\frac{1}{\nu}\right]_{c^2/a}^{a\Lambda^2+c^2/a} \\ &= \frac{c^2}{2\pi a}\left(\frac{a}{c^2} - \frac{1}{a\Lambda^2 + c^2/a}\right) = \frac{a\Lambda^2}{2\pi(a\Lambda^2 + c^2/a)}. \end{aligned} \tag{A.17}$$

In the $\Lambda \to \infty$ limit, $2\pi T \to 1$, matching the magnitude of the Chern number for either sector. For two normalized Bloch states, we define the quantum distance by $d^2(\mathbf{k},\mathbf{k}') \equiv 1 - |\langle u_-^\alpha(\mathbf{k}')|u_-^\alpha(\mathbf{k})\rangle|^2$. Taking $\mathbf{k}'=0$, the sector-independent distance follows from

$$|\langle u_-^\alpha(0)|u_-^\alpha(\mathbf{k})\rangle|^2 = \frac{c^2}{a^2k^2+c^2}, \qquad d^2(\mathbf{k},0) = 1 - \frac{c^2}{a^2k^2+c^2} = \frac{a^2k^2}{a^2k^2+c^2}, \tag{A.18}$$

where we used $|u_-^\alpha(0)\rangle = (-1,0)^T$ (from Eq. (A.5) at $\mathbf{k}=0$). At the cutoff edge $d_{\max} = \sqrt{a^2\Lambda^2/(a^2\Lambda^2+c^2)} = \sqrt{a\Lambda^2/(a\Lambda^2+c^2/a)} = \sqrt{2\pi T}$.

Appendix B: Landau-Level Spectrum Diagnostics

B.1. Exact Landau level spectrum

We provide a compact derivation of the Landau level (LL) energies for completeness. In the symmetric gauge $\mathbf{A} = \frac{\mathcal{B}}{2}(-y,x,0)$ and with $B = e\mathcal{B}/\hbar$, the kinetic momentum operators are

$$\Pi_x = -\mathrm{i}\partial_x - \frac{B}{2}y, \Pi_y = -\mathrm{i}\partial_y + \frac{B}{2}x, \qquad [\Pi_x,\Pi_y] = -\mathrm{i}B. \tag{B.1}$$

Defining $\Pi_\pm = \Pi_x \pm \mathrm{i}\Pi_y$, we have $[\Pi_-,\Pi_+] = 2B$. The cyclotron ladder operators $b = \Pi_-/\sqrt{2B}$, $b^\dagger = \Pi_+/\sqrt{2B}$ satisfy $[b,b^\dagger]=1$, and $\Pi_x^2+\Pi_y^2 = 2B(b^\dagger b + \frac{1}{2})$ so that $a(\Pi_x^2+\Pi_y^2) = aB(2\hat{n}+1)$ with $\hat{n} = b^\dagger b$. In this basis, the shifted Hamiltonians $\widetilde{H}_\alpha = H_\alpha - E_0\mathbb{I}$ take the form

$$\widetilde{H}_\uparrow = \begin{pmatrix} aB(2\hat{n}+1) & c\sqrt{2B}\,b^\dagger \\ c\sqrt{2B}\,b & c^2/a \end{pmatrix}, \qquad \widetilde{H}_\downarrow = \begin{pmatrix} aB(2\hat{n}+1) & -c\sqrt{2B}\,b \\ -c\sqrt{2B}\,b^\dagger & c^2/a \end{pmatrix}. \tag{B.2}$$

Spin-up ($H_\uparrow$) sector. The Hamiltonian $\widetilde{H}_\uparrow$ acts on the two-component spinor $|\Psi\rangle = (u|n,m\rangle,\, v|n-1,m\rangle)^T$ for $n \ge 1$. Using $\Pi_+|n-1,m\rangle = \sqrt{2Bn}|n,m\rangle$ and $\Pi_-|n,m\rangle = \sqrt{2Bn}|n-1,m\rangle$, the eigenvalue equation reduces to the 2×2 block

$$\begin{pmatrix} (2n+1)aB & c\sqrt{2Bn} \\ c\sqrt{2Bn} & c^2/a \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix} = (\lambda - E_0)\begin{pmatrix} u \\ v \end{pmatrix}. \tag{B.3}$$

The characteristic equation reads $\lambda'^2 - U_n\lambda' + c^2B = 0$ with $\lambda' = \lambda - E_0$ and $U_n = c^2/a + (2n+1)aB$, giving $\lambda' = \frac{1}{2}[U_n \pm \sqrt{U_n^2 - 4c^2B}]$, which yields Eq. (5) of the main text.

The $n=0$ zero mode for spin-up sector follows because $\Pi_-|0,m\rangle = 0$ annihilates the off-diagonal term and only $u|0,m\rangle$ survives with eigenvalue $E_0 + aB$, giving $E_0^\uparrow(B) = E_0 + aB$.

Spin-down ($H_\downarrow$) sector. For $\widetilde{H}_\downarrow$ (time-reversed sector), the natural spinor is $(\tilde{u}|n-1,m\rangle, \tilde{v}|n,m\rangle)^T$. After absorbing a relative minus sign into one spinor component, the eigenvalue equation reduces to the analogous block

$$\begin{pmatrix} (2n-1)aB & c\sqrt{2Bn} \\ c\sqrt{2Bn} & c^2/a \end{pmatrix}\begin{pmatrix} \tilde{u} \\ \tilde{v} \end{pmatrix} = (\lambda - E_0)\begin{pmatrix} \tilde{u} \\ \tilde{v} \end{pmatrix}. \tag{B.4}$$

with characteristic equation $\lambda'^2 - V_n\lambda' - c^2B = 0$ where $V_n = c^2/a + (2n-1)aB$. This gives $\lambda' = \frac{1}{2}[V_n \pm \sqrt{V_n^2 + 4c^2B}]$, yielding Eq. (6) of the main text. The $n=0$ zero mode for spin-down sector is $|\Psi\rangle = (0,|0,m\rangle)^T$ with $E_0^\downarrow = E_0 + c^2/a = E_1$ (field-independent).

B.2. LL Gap scaling asymptotics

Here we focus on the spin-up sector and discuss the gap scaling of the anomalous LL manifold $E_-^\uparrow(n)$ with $n = 1,\ldots,N_{\max}$, while the LL gap scaling for the spin-down sector $E_-^\downarrow(n)$ is similar. We distinguish the top gap, which involves the zero mode, from the gaps internal to the anomalous manifold:

$$G_0^\uparrow = E_0^\uparrow - E_-^\uparrow(1), \qquad G_n^\uparrow = E_-^\uparrow(n) - E_-^\uparrow(n+1), \quad n = 1,\ldots,N_{\max}-1. \tag{B.5}$$

For $U_n^2 \gg 4c^2B$ (valid when $B \ll B^* \equiv c^2/a^2$ or n is large), one expands the square root: $\sqrt{U_n^2 - 4c^2B} \approx U_n - 2c^2B/U_n$, giving

$$E_-^{\uparrow}(n) \approx E_0 + c^2B/U_n. \tag{B.6}$$

From Eq. (B.5), we have:

- **Internal gaps**: for adjacent anomalous levels in the small-B or large-n regime,

$$G_n^{\uparrow} \approx c^2B\left(\frac{1}{U_n} - \frac{1}{U_{n+1}}\right) = \frac{2ac^2B^2}{U_nU_{n+1}}, \qquad U_n = \frac{c^2}{a} + (2n+1)aB. \tag{B.7}$$

  For fixed n and $B \to 0$, this reduces to

$$G_n^{\uparrow} \approx 2a^3B^2/c^2. \tag{B.8}$$

- **Top gap** ($n = 0$): the exact gap between $E_0^{\uparrow}$ and $E_-^{\uparrow}(1)$ is

$$G_0^{\uparrow} = -\frac{1}{2}(aB + \frac{c^2}{a}) + \frac{1}{2}\sqrt{(\frac{c^2}{a})^2 + 2c^2B + 9a^2B^2}. \tag{B.9}$$

  (Derivation: $G_0^{\uparrow} = (E_0 + aB) - [E_0 + \frac{1}{2}(U_1 - \sqrt{U_1^2 - 4c^2B})]$ with $U_1 = c^2/a + 3aB$; one simplifies $(c^2/a + 3aB)^2 - 4c^2B = (c^2/a)^2 + 2c^2B + 9a^2B^2$.) For $B \ll B^*$ ($a^2B \ll c^2$), $G_0^{\uparrow} \approx 2a^3B^2/c^2$. For $B \gg B^*$ ($a^2B \gg c^2$), $G_0^{\uparrow} \approx aB - c^2/(3a)$.

For fixed n, the low-field limit is controlled by $(2n+1)aB \ll c^2/a$. In this regime the gaps defined in Eq. (B.5) are smooth functions of B and scale as B^2 according to Eq. (B.8), with an n-dependent crossover scale. For a fixed carrier density away from the continuum cutoff, the relevant spacing is obtained by evaluating Eq. (B.7) near the density-selected index $n^*(\rho, B)$. Fig. B.1 plots the LL gap spacings defined in Eq. (B.5).

Appendix C: Branch-Resolved LK Formulas

C.1. Poisson-summation and contour evaluation

We outline the derivation of Eq. (8) and Eq. (9) of the main text from the exact LL spectrum. For a LL branch labelled by spin sector α and band index $\eta = \pm$, with energies $E_\eta^\alpha(n, B)$ and degeneracy per unit area $D = B/(2\pi)$, the grand potential per unit area is

$$\Omega(B, T, \mu) = -k_BTD\sum_{\alpha,\eta}\sum_{n=0}^{N_{\max}} \ln\bigl[1 + e^{-(E_\eta^\alpha(n,B)-\mu)/k_BT}\bigr]. \tag{C.1}$$

Applying Poisson summation to this sum isolates the oscillatory part as

$$\begin{aligned}\delta\Omega(B, T, \mu) &= \sum_{\alpha,\eta}\delta\Omega_{\alpha,\eta}(B, T, \mu),\\ \delta\Omega_{\alpha,\eta}(B, T, \mu) &= -2k_BTD\,\Re\sum_{p=1}^{\infty}\int_{-1/2}^{N_{\max}+1/2} dn\,\ln\bigl[1 + e^{-(E_\eta^\alpha(n,B)-\mu)/k_BT}\bigr]\,e^{2\pi i pn},\end{aligned} \tag{C.2}$$

with $D = B/(2\pi)$ and $\Re$ takes the real part. Here the index n is treated as a continuous variable, and the upper limit $N_{\max}$ is set by the continuum cutoff. Starting from this form, change variable to $\varepsilon = E_\eta^\alpha(n, B)$ with $dn = d\varepsilon/v_\varepsilon$, where

$$v_\varepsilon = |\partial E_\eta^\alpha(n, B)/\partial n|_{n(\varepsilon)} \tag{C.3}$$

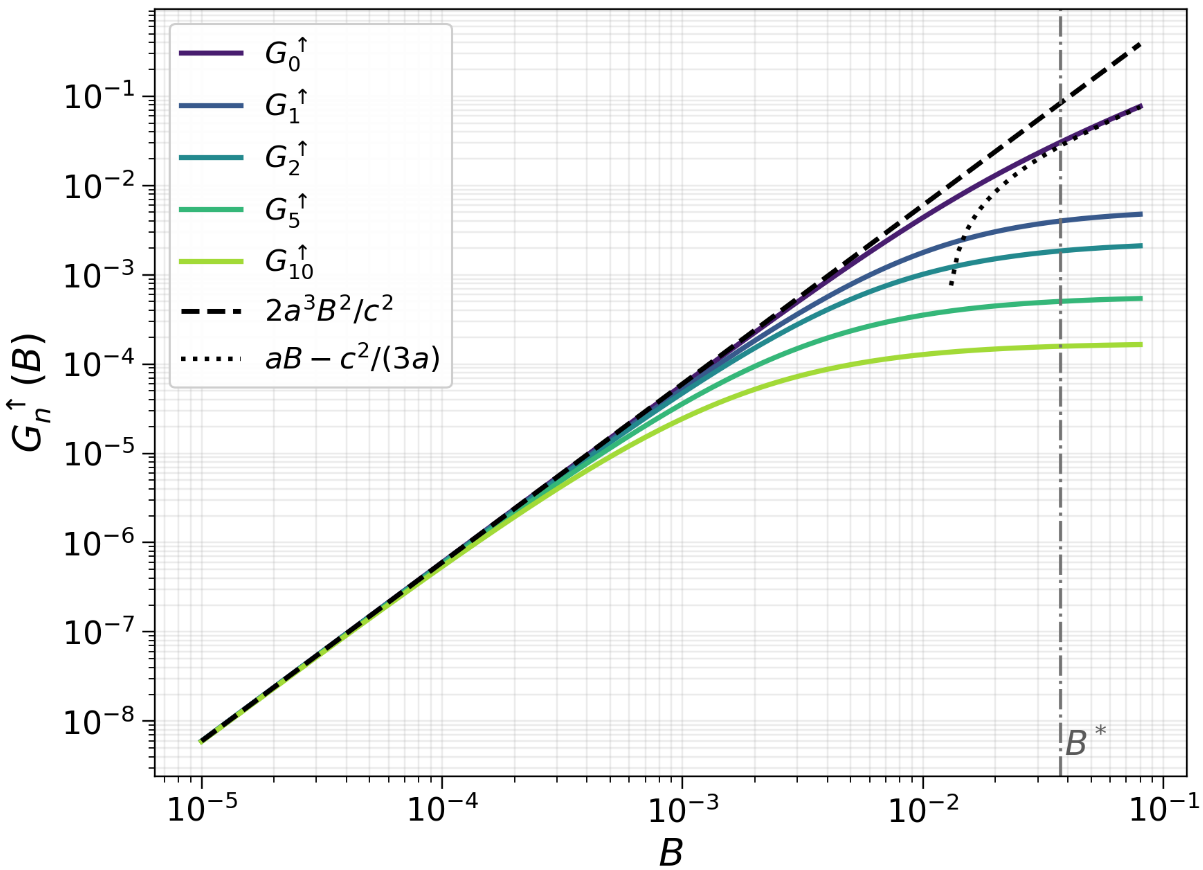

FIG. B.1. Exact fixed-index gap scaling from Eq. (B.5) for the spin-up anomalous LL manifold. The dashed line shows the fixed-index low-field asymptote $2a^3B^2/c^2$, while the dotted line shows the large-field asymptote $aB - c^2/(3a)$ for the top gap. The vertical dash-dotted line marks $B^* = c^2/a^2$.

is the local LL spacing. The temperature-dependent part of the integral is controlled by energies within a window of order k_BT around the chemical potential. When the local spacing varies slowly over this window, one may expand the inverse LL index $n_{\alpha,\eta}(\varepsilon, B)$ about the Fermi crossing $n^*_{\alpha,\eta}$ defined by

$$E^\alpha_\eta(n^*_{\alpha,\eta}, B) = \mu. \tag{C.4}$$

Keeping the leading term gives

$$n_{\alpha,\eta}(\varepsilon, B) \approx n^*_{\alpha,\eta} + \frac{\varepsilon - \mu}{v^{\alpha,\eta}_\mu}, \qquad v^{\alpha,\eta}_\mu \equiv \left|\partial E^\alpha_\eta(n, B)/\partial n\right|_{n^*_{\alpha,\eta}}, \tag{C.5}$$

where $v^{\alpha,\eta}_\mu$ is equivalent to the definition in Eq. (8) of the main text. Taking $v_\varepsilon \approx v^{\alpha,\eta}_\mu$ outside the integral gives

$$\delta\Omega_{\alpha,\eta} \approx -\frac{2k_BTD}{v^{\alpha,\eta}_\mu}\Re\sum_{p=1}^{\infty} e^{2\pi i p n^*_{\alpha,\eta}} \int d\varepsilon \ln\!\left[1 + e^{-(\varepsilon-\mu)/k_BT}\right] e^{2\pi i p(\varepsilon-\mu)/v^{\alpha,\eta}_\mu}. \tag{C.6}$$

Due to the logarithm in Eq. (C.6), we first need to consider the integral boundary. For $\xi = (\varepsilon - \mu)/(k_BT) \to -\infty$, $\ln(1 + e^{-\xi}) \sim -\xi$, so the boundary term from a direct integration by parts would not vanish. Let $F(\xi) = \ln(1 + e^{-\xi})$, $\tilde{v} = 2\pi p k_BT/v^{\alpha,\eta}_\mu$, and keep the finite dimensionless endpoints $\xi_\pm = (\varepsilon_\pm - \mu)/(k_BT)$

inherited from the LL cutoff. For this finite interval, two integrations by parts give the exact identity

$$\int_{\xi_-}^{\xi_+} d\xi\, F(\xi) e^{\mathrm{i}\tilde{v}\xi} = \mathcal{B}_{\mathrm{end}} - \frac{1}{\tilde{v}^2} \int_{\xi_-}^{\xi_+} d\xi\, F''(\xi) e^{\mathrm{i}\tilde{v}\xi}, \qquad F''(\xi) = \frac{1}{4}\,\mathrm{sech}^2\!\left(\frac{\xi}{2}\right). \tag{C.7}$$

Here $\mathcal{B}_{\mathrm{end}}$ contains only finite-endpoint terms evaluated at $\xi_\pm$. These terms are cutoff/edge contributions, whereas the LK oscillation associated with the Fermi crossing is controlled by the nonanalytic change of occupation at $\varepsilon = \mu$ at zero temperature. When the Fermi level is far from the endpoints on the scale of $k_B T$, the Fermi-surface contribution can therefore be isolated by extending the localized kernel $F''(\xi)$ to the full real line. In the $T \to 0$ limit, $F''(\xi)$ tends to $\delta(\xi)$, giving the undamped $T = 0$ harmonic; at finite T the broadened kernel gives the LK damping factor

$$R_T(\tilde{v}) = \int_{-\infty}^{\infty} d\xi\, \left[-f_0'(\xi)\right] e^{\mathrm{i}\tilde{v}\xi} = \int_{-\infty}^{\infty} \frac{d\xi}{4}\,\mathrm{sech}^2\!\left(\frac{\xi}{2}\right) e^{\mathrm{i}\tilde{v}\xi}, \tag{C.8}$$

where $f_0(\xi) = (1 + e^{\xi})^{-1}$. In this expression for $R_T(\tilde{v})$, the imaginary part vanishes because the thermal kernel is even in ξ, whereas $\sin(\tilde{v}\xi)$ is odd. Closing the contour in the upper half-plane picks up the Matsubara poles of $f_0'(\xi)$ at $\xi_k = \mathrm{i}\pi(2k+1)$ $(k = 0, 1, 2, \ldots)$:

$$R_T(\tilde{v}) = \frac{X_p^{\alpha,\eta}}{\sinh(X_p^{\alpha,\eta})}. \tag{C.9}$$

where

$$X_p^{\alpha,\eta} \equiv \pi\tilde{v} = 2\pi^2 p k_B T / v_\mu^{\alpha,\eta}. \tag{C.10}$$

Collecting all prefactors – the Jacobian $1/v_\mu^{\alpha,\eta}$ from the change of variable, two integration-by-parts factors of $v_\mu^{\alpha,\eta}/(2\pi p)$, and the explicit D factor – yields

$$\delta\Omega_{\alpha,\eta}(B, T, \mu) = \frac{D}{2\pi^2} \sum_{p=1}^{\infty} \frac{v_\mu^{\alpha,\eta}}{p^2} \frac{X_p^{\alpha,\eta}}{\sinh X_p^{\alpha,\eta}} \cos\!\big[2\pi p\, n^*_{\alpha,\eta}(\mu, B)\big], \tag{C.11}$$

in agreement with Eq. (8) of the main text. (For complete bookkeeping, see e.g. Shoenberg, *Magnetic Oscillations in Metals*, §2.3, eqs. 2.151–2.157.)

C.2. Local LL spacing for each branch

$E_-^{\uparrow}$ branch ($\mu > E_0$, wedge condition $0 < \delta\mu < c\sqrt{B}$). Setting $E_-^{\uparrow}(n^*, B) = \mu$ with $\delta\mu = \mu - E_0 > 0$ means

$$2\delta\mu = U_{n^*} - \sqrt{U_{n^*}^2 - 4c^2 B}. \tag{C.12}$$

Rearranging to isolate the square root and squaring: $(U_{n^*} - 2\delta\mu)^2 = U_{n^*}^2 - 4c^2 B$, which gives $4\delta\mu^2 - 4\delta\mu U_{n^*} = -4c^2 B$, hence

$$U_{n^*} \equiv \frac{c^2}{a} + (2n^* + 1)aB = \delta\mu + \frac{c^2 B}{\delta\mu}. \tag{C.13}$$

(The squaring step requires $U_{n^*} > 2\delta\mu$, i.e. $c^2 B > \delta\mu^2$, which is precisely the wedge condition.) Solving Eq. (C.13) for n^*:

$$n^* + \frac{1}{2} = \frac{U_{n^*} - c^2/a}{2aB} = -\frac{E_0 + c^2/a - \mu}{2aB} + \frac{c^2}{2a(\mu - E_0)}. \tag{C.14}$$

Here the combination $E_0 + c^2/a$ enters only through the ideal-flat-band offset in the spin-up block. The local spacing follows from $\partial E_-/\partial n\big|_{n^*}$. Differentiating $E_-^{\uparrow} = E_0 + \frac{1}{2}(U_n - \sqrt{U_n^2 - 4c^2B})$ with respect to n (using $\partial U_n/\partial n = 2aB$) and evaluating at n^* where $\sqrt{U_{n^*}^2 - 4c^2B} = U_{n^*} - 2\delta\mu$:

$$\frac{\partial E_-^{\uparrow}}{\partial n}\bigg|_{n^*} = aB\left(1 - \frac{U_{n^*}}{U_{n^*} - 2\delta\mu}\right) = \frac{-2aB\,\delta\mu}{U_{n^*} - 2\delta\mu} = \frac{-2aB\,\delta\mu^2}{c^2B - \delta\mu^2}, \tag{C.15}$$

where we used $U_{n^*} - 2\delta\mu = (c^2B - \delta\mu^2)/\delta\mu$ from Eq. (C.13). Since $\delta\mu < c\sqrt{B}$, the denominator $c^2B - \delta\mu^2 > 0$ and the slope is negative (levels decrease in n). Taking the absolute value of this local slope gives the positive LL spacing:

$$v_\mu^{\uparrow,-} \equiv \left|\frac{\partial E_-^{\uparrow}}{\partial n}\right|_{n^*} = \frac{2aB(\mu - E_0)^2}{c^2B - (\mu - E_0)^2}. \tag{C.16}$$

$E_-^{\downarrow}$ branch ($\mu < E_0$, valid for $-c\sqrt{B} < \delta\mu < 0$). Setting $E_-^{\downarrow}(n^*, B) = \mu$ with $\delta\mu = \mu - E_0 < 0$:

$$2\delta\mu = V_{n^*} - \sqrt{V_{n^*}^2 + 4c^2B}. \tag{C.17}$$

Isolating the square root and squaring: $(V_{n^*} - 2\delta\mu)^2 = V_{n^*}^2 + 4c^2B$, giving $4\delta\mu^2 - 4\delta\mu V_{n^*} = 4c^2B$, hence (with $\delta\mu < 0$)

$$V_{n^*} \equiv \frac{c^2}{a} + (2n^* - 1)aB = \delta\mu - \frac{c^2B}{\delta\mu}. \tag{C.18}$$

Solving for n^*:

$$n^* - \frac{1}{2} = \frac{V_{n^*} - c^2/a}{2aB} = \frac{c^2}{2a(E_0 - \mu)} - \frac{E_0 + c^2/a - \mu}{2aB}. \tag{C.19}$$

The local spacing can be obtained directly from the slope of the spin-down anomalous branch. Differentiating $E_-^{\downarrow} = E_0 + \frac{1}{2}(V_n - \sqrt{V_n^2 + 4c^2B})$ with respect to n and using $\partial V_n/\partial n = 2aB$ gives

$$\frac{\partial E_-^{\downarrow}}{\partial n}\bigg|_{n^*} = aB\left(1 - \frac{V_{n^*}}{\sqrt{V_{n^*}^2 + 4c^2B}}\right). \tag{C.20}$$

At the Fermi crossing, Eq. (C.17) gives $\sqrt{V_{n^*}^2 + 4c^2B} = V_{n^*} - 2\delta\mu$. Using Eq. (C.18),

$$V_{n^*} - 2\delta\mu = -\delta\mu - \frac{c^2B}{\delta\mu} = -\frac{\delta\mu^2 + c^2B}{\delta\mu}, \tag{C.21}$$

which is positive because $\delta\mu < 0$. Therefore

$$\frac{\partial E_-^{\downarrow}}{\partial n}\bigg|_{n^*} = aB\left[1 + \frac{\delta\mu^2 - c^2B}{\delta\mu^2 + c^2B}\right] = \frac{2aB\delta\mu^2}{\delta\mu^2 + c^2B}. \tag{C.22}$$

The slope is positive for this branch, so the positive local spacing is

$$v_\mu^{\downarrow,-} \equiv \left|\frac{\partial E_-^{\downarrow}}{\partial n}\right|_{n^*} = \frac{2aB(\mu - E_0)^2}{(\mu - E_0)^2 + c^2B}. \tag{C.23}$$

Semiclassical (bulk) limit and connection to quantum geometry. In the bulk of the anomalous manifold ($|\delta\mu| \ll c\sqrt{B}$, large n^*), Eq. (C.16) first reduces to

$$v_\mu^{\uparrow,-} = \frac{2aB\delta\mu^2}{c^2B - \delta\mu^2} \approx \frac{2a\delta\mu^2}{c^2}. \tag{C.24}$$

To relate $\delta\mu$ to the zero-field band geometry, use the large-n^* semiclassical identification $k_{n^*}^2 \approx 2n^*B$. In the same regime the anomalous LL energy follows from expanding the lower branch as

$$\delta\mu = E_-^{\uparrow}(n^*, B) - E_0 \approx \frac{c^2B}{U_{n^*}} \approx \frac{c^2B}{\Delta(k_{n^*})}, \qquad \Delta(k) = ak^2 + \frac{c^2}{a}, \tag{C.25}$$

where we have used Eq.(B.6). Substituting this estimate for $\delta\mu$ into the bulk expression for v_μ gives

$$v_\mu^{\uparrow,-} \approx \frac{2a\delta\mu^2}{c^2} \approx \frac{2ac^2B^2}{\Delta(k_{n^*})^2} = aB^2 \operatorname{tr} g(k_{n^*}), \tag{C.26}$$

where the last equality uses $\operatorname{tr} g = 2c^2/\Delta^2$ from Eq. (A.14). The same relation holds for the $E_-^{\downarrow}$ branch. Thus the local LL spacing directly encodes the quantum metric of the flat band, establishing $v_\mu^{\alpha,-} \sim aB^2 \operatorname{tr} g$ as the fundamental connection between anomalous-LL thermal damping and quantum geometry.

Upper branches ($E_+^{\uparrow}$, $E_+^{\downarrow}$). For the upper branches, the same differentiation of the exact LL energies gives

$$v_\mu^{\uparrow,+} = \frac{2aB(\mu - E_0)^2}{(\mu - E_0)^2 - c^2B}, \qquad v_\mu^{\downarrow,+} = \frac{2aB(\mu - E_0)^2}{(\mu - E_0)^2 + c^2B}. \tag{C.27}$$

In the dispersive regime $|\mu - E_0| \gg c\sqrt{B}$, both upper-branch spacings reduce to $v_\mu^{\alpha,+} \approx 2aB$, recovering the standard parabolic-band LK factor $X_p^{\alpha,+} = \pi^2 p k_B T/(aB)$ with effective mass $m^*_{\text{eff},\alpha+} = 1/(2a)$.

C.3. Effective mass formulas

Using $m^*_{\text{eff}} = B/v_\mu$ and the above expressions for v_μ gives:

$$m^*_{\text{eff},\uparrow-} = \frac{c^2B - (\mu - E_0)^2}{2a(\mu - E_0)^2}, \tag{C.28}$$

$$m^*_{\text{eff},\downarrow-} = \frac{1}{2a} + \frac{c^2B}{2a(\mu - E_0)^2}. \tag{C.29}$$

In the anomalous bulk regime, $\delta\mu^2 \ll c^2B$, the two lower-branch masses have the same leading behavior,

$$m^*_{\text{eff},\uparrow-} \simeq m^*_{\text{eff},\downarrow-} \simeq \frac{c^2B}{2a(\mu - E_0)^2}, \tag{C.30}$$

up to the subleading additive constants $\mp 1/(2a)$. This strong field dependence at fixed μ contrasts with the constant parabolic value $m^*_{\text{eff},\alpha+} = 1/(2a)$ of the dispersive upper branches. At fixed carrier density, however, the field dependence must be evaluated along the self-consistent trajectory $\mu_\rho(B)$. For a single active lower-branch sequence, the LL degeneracy is $D = B/(2\pi)$, so fixed active density $\rho_{\mathcal{S}}$ (defined in Sec. D as the density counted by the active sector) selects

$$n^*(B) \simeq \frac{|\rho_{\mathcal{S}}|}{D} = \frac{2\pi|\rho_{\mathcal{S}}|}{B}. \tag{C.31}$$

The semiclassical momentum tied to this crossing is therefore

$$k_{n^*}^2 \simeq 2n^*B \simeq 4\pi|\rho_{\mathcal{S}}|, \tag{C.32}$$

which is approximately independent of B at fixed density. Using the weak-field anomalous-branch dispersion,

$$\mu_\rho(B) - E_0 \simeq s_\alpha \frac{c^2B}{c^2/a + ak_{n^*}^2} \simeq s_\alpha \frac{c^2}{c^2/a + 4\pi a|\rho_{\mathcal{S}}|} B, \qquad s_\uparrow = +1, \quad s_\downarrow = -1, \tag{C.33}$$

so the chemical-potential offset from the flat-band energy is linear in B along the fixed-density trajectory. Substituting Eq. (C.33) into Eq. (C.30) gives

$$m^*_{\text{eff},\alpha-}\big(B, \mu_\rho(B)\big) \simeq \frac{\big(c^2/a + 4\pi a|\rho_{\mathcal{S}}|\big)^2}{2ac^2} \frac{1}{B} = \frac{1}{aB \operatorname{tr} g_\alpha(k_{n^*})}, \tag{C.34}$$

which is the roughly $1/B$ trend shown by the dashed theory curves in Fig. 4 of the main text.

Appendix D: Numerical Results for Fixed-Density Thermodynamics

D.1. Density convention and Fermi level

We first fix the density convention used in all numerical plots. The signed carrier density ρ is measured relative to the midpoint of the two zero modes,

$$E_{\rm ref}(B) = \frac{E_0^{\uparrow}(B) + E_0^{\downarrow}(B)}{2}. \tag{D.1}$$

Thus $\rho > 0$ denotes electron doping above $E_{\rm ref}$, while $\rho < 0$ denotes hole doping below $E_{\rm ref}$. The corresponding discrete LL spectral density per unit area is

$$\nu_{\rm LL}(E,B) = \frac{B}{2\pi}\left[\sum_{\alpha=\uparrow,\downarrow}\delta(E - E_0^{\alpha}(B)) + \sum_{\alpha=\uparrow,\downarrow}\sum_{\eta=\pm}\sum_{n=1}^{N_{\max}}\delta\big(E - E_\eta^{\alpha}(n,B)\big)\right], \tag{D.2}$$

where $E_0^{\alpha}(B)$ denotes the zero mode in spin sector α, and $E_\eta^{\alpha}(n,B)$ denotes the branch-resolved LL energy introduced in Appendix B. At finite temperature, the signed density associated with a chemical potential μ is obtained by integrating the LL spectral density with the Fermi occupation,

$$\rho(\mu,B,T) = \int dE\, \nu_{\rm LL}(E,B)\left[f_T(E-\mu) - \Theta(E_{\rm ref}(B) - E)\right], \qquad f_T(x) = \frac{1}{e^{x/k_BT}+1}. \tag{D.3}$$

The second term subtracts the reference filling with all LLs below $E_{\rm ref}$ occupied and all LLs above $E_{\rm ref}$ empty. For a prescribed carrier density, the chemical potential $\mu_\rho(B,T)$ is determined by solving $\rho = \rho(\mu_\rho, B, T)$. In the $T \to 0$ limit, Eq. (D.3) reduces to the usual signed LL-counting convention relative to $E_{\rm ref}$.

Since we need to choose a momentum cutoff Λ for the topological flat bands, which also introduces a cutoff in the anomalous LL spectrum, it is more stable to work with the active carrier sector rather than with the full signed integral each time. This gives three cases:

$$\begin{aligned}
\rho > 0: &\quad \rho_{\mathcal{S}} = \rho, && \mathcal{S} = \{\text{electron LLs above } E_{\rm ref}\},\\
-\rho_{\rm cap}/2 < \rho < 0: &\quad \rho_{\mathcal{S}} = |\rho|, && \mathcal{S} = \{\text{hole levels counted downward from } E_{\rm ref}\},\\
\rho < -\rho_{\rm cap}/2: &\quad \rho_{\mathcal{S}} = \rho_{{\rm low},e} = \rho_{\rm cap} - |\rho|, && \mathcal{S} = \{\text{remaining electrons in the lower manifold}\}.
\end{aligned} \tag{D.4}$$

The shallow-hole description is used when the missing carriers are still between E_0 and $E_{\rm ref}$, so the chemical potential is within the upper manifold of anomalous LLs. The maximal carrier density of the two anomalous sectors together is

$$\rho_{\rm cap} = \frac{\Lambda^2}{2\pi}. \tag{D.5}$$

Thus the shallow/deep boundary in Eq. (D.4) occurs at $\rho_{\rm cap}/2 = \Lambda^2/(4\pi)$. For $\Lambda = 1.0\,{\rm nm}^{-1}$, one flat band has capacity $0.0796\,{\rm nm}^{-2}$, while the combined two-sector capacity is $\rho_{\rm cap} = 0.159\,{\rm nm}^{-2}$; therefore $\rho = -0.140$ and $-0.150\,{\rm nm}^{-2}$ are deep-hole examples used below for numerical calculations in this combined convention. This rewriting is only a carrier-counting convention: the oscillations are then controlled by the small number $\rho_{{\rm low},e}$ of lower-manifold electrons left unremoved. Consequently, the black reference lines in the FFT plots use $F = 2\pi|\rho|$ for the electron and shallow-hole sectors, whereas the deep-hole reference uses $F = 2\pi\rho_{{\rm low},e}$. As discussed below, the $\rho = +0.020\,{\rm nm}^{-2}$ magnetization spectrum also contains a dominant low-frequency peak near $F = \pi\rho$. This feature correlates with the near-Fermi LL splitting structure and does not alter the fixed-density counting convention.

For each case in Eq. (D.4), we introduce an active-sector energy variable $\varepsilon_\ell^{\mathcal{S}}(B)$. Let $E_\ell(B)$ denote the LL energies $E_0^{\alpha}(B)$ and $E_\eta^{\alpha}(n,B)$ in the original electron spectrum. Then

$$\varepsilon_\ell^{\mathcal{S}}(B) = \begin{cases} E_\ell(B), & \mathcal{S} = \{\text{electron LLs above } E_{\rm ref}\},\\ E_{\rm ref}(B) - E_\ell(B), & \mathcal{S} = \{\text{hole levels counted downward from } E_{\rm ref}\},\\ E_\ell(B), & \mathcal{S} = \{\text{remaining electrons in the lower manifold}\}. \end{cases} \tag{D.6}$$

The sector density $\rho_{\mathcal{S}}$ is therefore the active carrier density for the chosen sector. Only after this sector energy variable is fixed do we define the sector chemical potential $\mu_{\mathcal{S}}$. It is the quantity conjugate to $\rho_{\mathcal{S}}$, measured in the same energy variable as $\varepsilon_\ell^{\mathcal{S}}$, and is defined implicitly by

$$\rho_{\mathcal{S}} = D \sum_{\ell \in \mathcal{S}} f_T\left(\varepsilon_\ell^{\mathcal{S}}(B) - \mu_{\mathcal{S}}\right), \qquad D = \frac{B}{2\pi}, \tag{D.7}$$

which is the sector version of Eq. (D.3). For the electron and deep-hole sectors, $\mu_{\mathcal{S}}$ is the physical electron chemical potential restricted to the retained sector. For the shallow-hole sector, $\mu_{\mathcal{S}}$ is instead the hole chemical potential in the transformed variable $E_{\text{ref}} - E_\ell$, not the original electron chemical potential μ.

D.2. Fixed-density magnetization

For each active carrier sector $\mathcal{S}$, the sector grand potential is

$$\Omega_{\mathcal{S}}(B, T, \mu_{\mathcal{S}}) = -k_B T\, D \sum_{\ell \in \mathcal{S}} \ln\left[1 + \exp\!\left(-\frac{\varepsilon_\ell^{\mathcal{S}}(B) - \mu_{\mathcal{S}}}{k_B T}\right)\right], \tag{D.8}$$

Here $\mathcal{S}$, $\varepsilon_\ell^{\mathcal{S}}(B)$, $\rho_{\mathcal{S}}$, $\mu_{\mathcal{S}}(B, T, \rho_{\mathcal{S}})$, and D are the sector quantities defined above. The magnetization (per unit area) at fixed sector density is obtained from the Helmholtz free energy $f_{\mathcal{S}} = \Omega_{\mathcal{S}} + \mu_{\mathcal{S}}\rho_{\mathcal{S}}$:

$$M_{\mathcal{S}}(B, T, \rho_{\mathcal{S}}) = -\left.\frac{\partial f_{\mathcal{S}}}{\partial B}\right|_{T, \rho_{\mathcal{S}}}. \tag{D.9}$$

To see how the fixed-density derivative is evaluated, write the chemical potential on the fixed-density trajectory as $\mu_{\mathcal{S}}(B)$, suppressing T and $\rho_{\mathcal{S}}$ in the notation. Then

$$\begin{aligned}\left.\frac{df_{\mathcal{S}}}{dB}\right|_{T,\rho_{\mathcal{S}}} &= \left(\frac{\partial \Omega_{\mathcal{S}}}{\partial B}\right)_{T,\mu_{\mathcal{S}}} + \left(\frac{\partial \Omega_{\mathcal{S}}}{\partial \mu_{\mathcal{S}}}\right)_{T,B} \frac{d\mu_{\mathcal{S}}}{dB} + \rho_{\mathcal{S}} \frac{d\mu_{\mathcal{S}}}{dB} \\ &= \left(\frac{\partial \Omega_{\mathcal{S}}}{\partial B}\right)_{T,\mu_{\mathcal{S}}},\end{aligned} \tag{D.10}$$

where the last equality uses the thermodynamic identity $\rho_{\mathcal{S}} = -(\partial\Omega_{\mathcal{S}}/\partial\mu_{\mathcal{S}})_{T,B}$. Thus $M_{\mathcal{S}}$ can equivalently be computed as $-(\partial\Omega_{\mathcal{S}}/\partial B)_{T,\mu_{\mathcal{S}}}$ evaluated at $\mu_{\mathcal{S}} = \mu_{\mathcal{S}}(B, T, \rho_{\mathcal{S}})$. In the numerical evaluation, the B-derivative in Eq. (D.10) can be obtained analytically from the explicit LL energies $E_0^\alpha(B)$ and $E_\eta^\alpha(n, B)$ given in Appendix B.

The oscillatory part $\widetilde{M}$ is obtained numerically by subtracting a smooth background in $1/B$. For the anomalous densities, we adopt the M_{no0LL} convention: the zero-mode contributions are excluded from the oscillatory signal to avoid contamination by the field-independent $E_0^\downarrow$ and the trivial B-linear $E_0^\uparrow$ shift, which has no contribution to oscillation behaviors. We also evaluate the exact discrete sector grand potential in Eq. (D.8) along the fixed-density trajectory and subtract the same type of smooth background, denoting the resulting oscillatory component by $\delta\Omega_\rho$. For deep-hole densities, the active density is the remaining lower-manifold electron density defined in Eq. (D.4). This construction is shown directly in Fig. D.1, which plots the resulting thermodynamic oscillations over a wide $1/B$ range for the six densities used in the appendix. The left column shows the background-subtracted grand potential $\delta\Omega_\rho$ obtained from the exact LL sum, while the right column shows the background-subtracted, zero-mode-excluded magnetization $\widetilde{M}_{\text{no0LL}}$. Thus Fig. D.1 serves as the starting point for the subsequent FFT, window, and thermal-damping analyses, and it also makes the temperature dependence visible before any Fourier or local-window processing is applied. At the lowest temperatures, the electron densities in the dispersive regime, Figs. D.1(a)–D.1(d), show many well-resolved oscillation periods across the plotted $1/B$ range, whereas the hole densities in the anomalous regime, Figs. D.1(e)–D.1(l), lose amplitude rapidly toward smaller B (larger $1/B$), leaving only a few visible periods before the oscillatory signal becomes strongly suppressed.

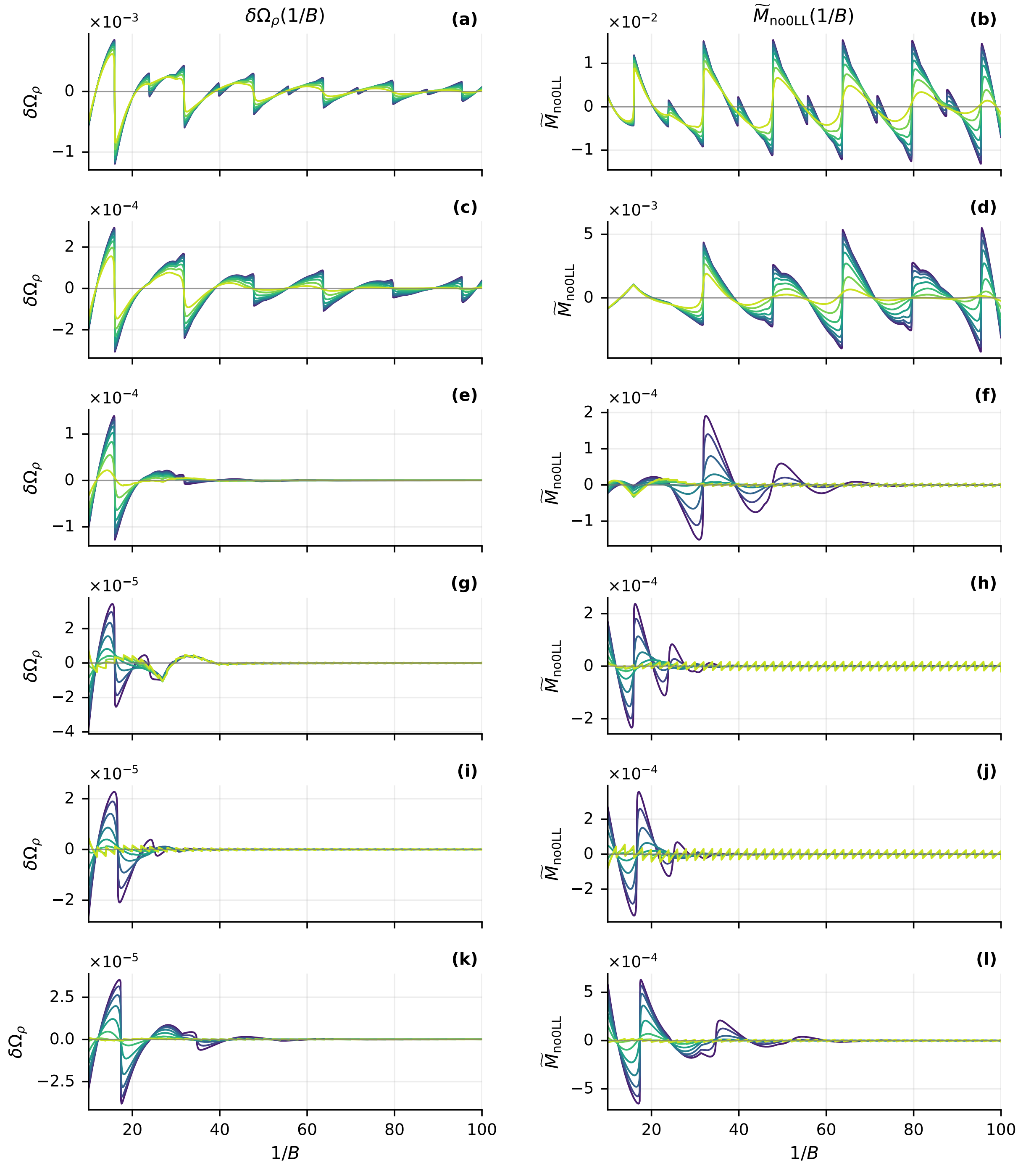

FIG. D.1. Wide-range thermodynamic oscillations at $\Lambda = 1.0\,\mathrm{nm}^{-1}$ for $1/B \in [10, 100]$. In each panel, the curves correspond to $k_BT = (2.00,\ 3.07,\ 4.71,\ 7.22,\ 11.1,\ 17.0,\ 26.1,\ 40.0) \times 10^{-4}\,\mathrm{eV}$, ordered from purple at the lowest temperature to yellow at the highest temperature. The left column shows the background-subtracted sector grand potential $\delta\Omega_\rho(1/B)$ computed from the exact discrete LL sum in Eq. (D.8); the right column shows the background-subtracted, zero-mode-excluded oscillatory magnetization $\widetilde{M}_{\mathrm{no0LL}}(1/B)$ computed along the same fixed-density trajectory. The density assignments are (a,b) $\rho = +0.020$, (c,d) $\rho = +0.010$, (e,f) $\rho = -0.010$, (g,h) $\rho = -0.020$, (i,j) $\rho = -0.140$, and (k,l) $\rho = -0.150$, all in units of nm^{-2}.

D.3. FFT frequency diagnostics

Figure D.2 is the frequency-space diagnostic of the wide-range oscillations in Fig. D.1. For each density, we Fourier transform the lowest-temperature traces, $k_BT = 2.0 \times 10^{-4}\,\mathrm{eV}$, using the same $1/B$ interval as in Fig. D.1. The left column analyzes the background-subtracted grand potential $\delta\Omega_\rho$, while the right column analyzes the background-subtracted, zero-mode-excluded magnetization $\widetilde{M}_{\mathrm{no0LL}}$. The vertical dashed lines in Fig. D.2 are LL-counting reference frequencies used to interpret where the numerical FFT weight should appear, as discussed below.

For one set of LL ladder, the degeneracy per unit area is $D = B/(2\pi)$. If n_f levels are filled, the active carrier density is $\rho_{\mathcal{S}} = Dn_f$, so $n_f = (2\pi\rho_{\mathcal{S}})/B$. One oscillation occurs when n_f changes by one as a function of $1/B$, giving the fundamental frequency

$$F_{\rho_{\mathcal{S}}} = 2\pi\rho_{\mathcal{S}}. \tag{D.11}$$

Here $\rho_{\mathcal{S}}$ is the density counted by the active LL sequence. For the electron and shallow-hole rows, $\rho_{\mathcal{S}} = |\rho|$. For the deep-hole rows, the useful active variable is the remaining lower-manifold electron density $\rho_{\mathcal{S}} = \rho_{\mathrm{low},e}$ defined in Eq. (D.4), because the oscillations count the residual unfilled part of the lower anomalous manifold rather than the full hole density. The black dashed lines in Fig. D.2 mark $F = 2\pi\rho_{\mathcal{S}}$ for both electron and hole sectors. With this convention, the FFT spectra test whether the exact fixed-density LL calculation oscillates with the period expected from the active carrier density. The electron-side spectra are shown first in Figs. D.2(a)–D.2(d). For $\rho = +0.010\,\mathrm{nm}^{-2}$, panels (c) and (d) have their dominant peak consistent with the black line from Eq. (D.11), so the oscillations are well described by a spin-resolved LL ladder with $\rho_{\mathcal{S}} = \rho$. For $\rho = +0.020\,\mathrm{nm}^{-2}$, panels (a) and (b) instead show a stronger low-frequency peak described by the green dashed line $F = \pi\rho$, rather than by the black dashed line at $F = 2\pi\rho$. This additional peak is consistent with an effectively spin-degenerate density-frequency relation. If two spin ladders are sufficiently close over the relevant field interval, the fixed total carrier density is shared between them. If g_s spin ladders contribute with the same frequency, each ladder carries ρ/g_s, and Eq. (D.11) gives

$$F = 2\pi\frac{\rho}{g_s}. \tag{D.12}$$

Thus an effectively spin-degenerate electron sequence with $g_s = 2$ gives $F = \pi\rho$, motivating the green guide in panels (a) and (b). The FFT alone does not establish a unique degeneracy assignment. It instead indicates that the $\rho = +0.010\,\mathrm{nm}^{-2}$ trace is closer to the spin-resolved counting scale, whereas the $\rho = +0.020\,\mathrm{nm}^{-2}$ trace contains a strong spin-degenerate-like component. We therefore examine the fixed-density chemical potential and the nearest-Fermi LL spacing in Fig. D.3 as a direct LL-level diagnostic of the different electron-side spectra in Figs. D.2(a)–D.2(d). Panel (a) shows that the fixed-density chemical potential is determined from the same full spin-sector LL spectrum for both $\rho = +0.010$ and $+0.020\,\mathrm{nm}^{-2}$. Panel (b) then compares the nearest-Fermi-energy LL spacing ΔE_{near}, defined by the two LLs immediately below and above $\mu_{\mathcal{S}}$. For $\rho = +0.010\,\mathrm{nm}^{-2}$, the near-Fermi spacing is comparatively regular, consistent with the dominant FFT weight near the spin-resolved counting scale $F = 2\pi\rho$. For $\rho = +0.020\,\mathrm{nm}^{-2}$, ΔE_{near} has a much stronger oscillatory modulation over the same $1/B$ range, alternating between small spin splittings and larger inter-LL spacings. The corresponding chemical-potential modulation is visible in panel (a) of Fig. D.3. Together, these observations support—without uniquely proving—the spin-degenerate-like interpretation of the low-frequency FFT component.

On the hole side, the anomalous LLs from the two spin sectors are split into branches on opposite sides of E_0, so they do not form the same kind of nearly spin-degenerate sequence that appears for the electron-side dispersive LLs. The FFT spectra in Figs. D.2(e)–D.2(l) therefore generally show one main counting peak. Compared with the electron-side spectra, however, the hole-side peaks are much broader. For $\rho = -0.020$ and $-0.140\,\mathrm{nm}^{-2}$, the $\delta\Omega_\rho$ spectra in panels (g) and (i) are so broad that the peak position is difficult to assign reliably. This is consistent with the direct $1/B$-domain traces in Fig. D.1(g,i), where the $\delta\Omega_\rho$ oscillations decay rapidly as $1/B$ increases. The corresponding magnetization spectra in Figs. D.2(h) and D.2(j) still show visible peaks, but these peaks remain broader. For $\rho = -0.010$ and $-0.150\,\mathrm{nm}^{-2}$, the main FFT peaks can be resolved in both $\delta\Omega_\rho$ and $\widetilde{M}_{\mathrm{no0LL}}$, and their positions are consistent with the active-density counting frequency $F = 2\pi\rho_{\mathcal{S}}$ in Eq. (D.11), shown by the black dashed lines.

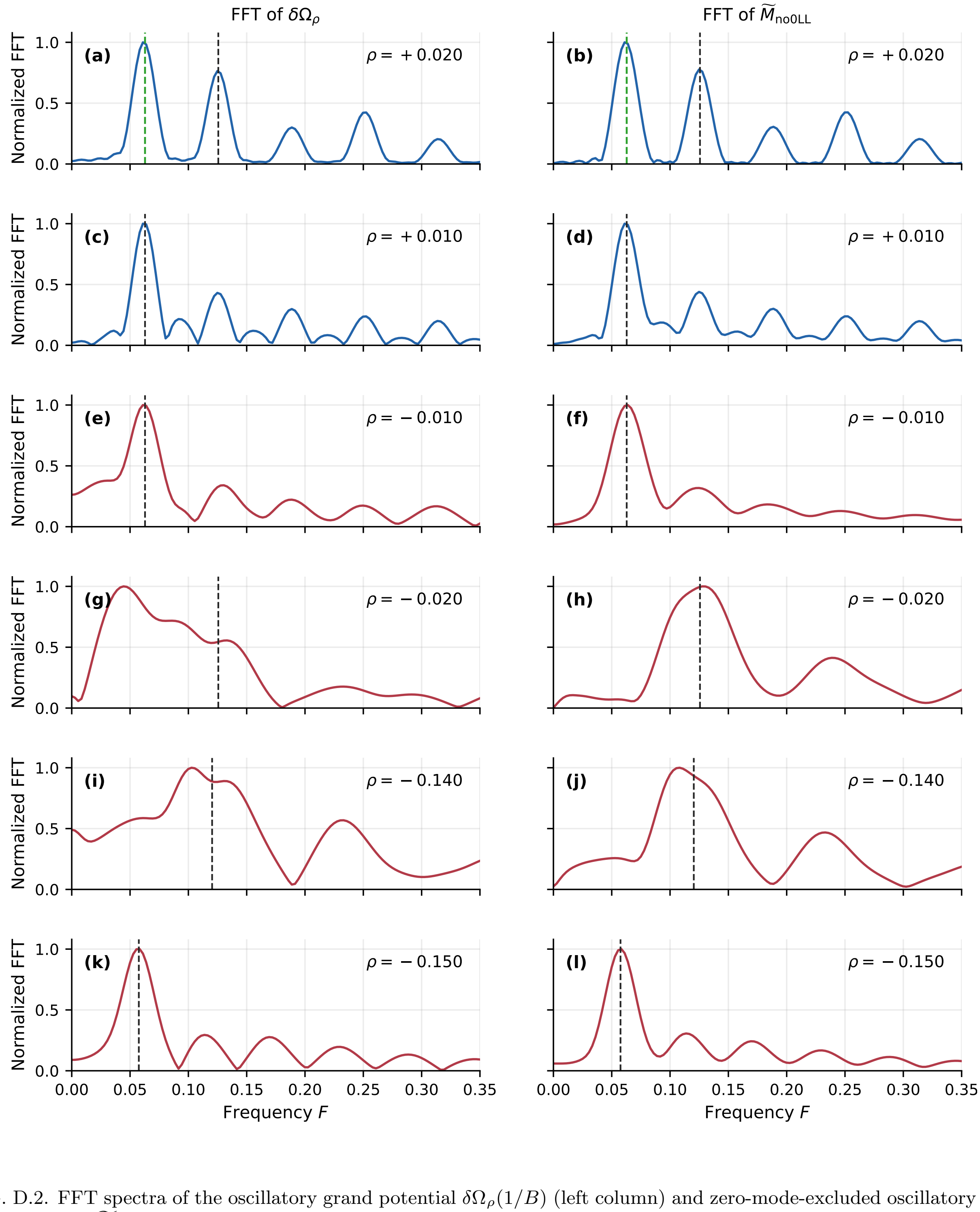

FIG. D.2. FFT spectra of the oscillatory grand potential $\delta\Omega_\rho(1/B)$ (left column) and zero-mode-excluded oscillatory magnetization $\widetilde{M}_{\mathrm{no0LL}}(1/B)$ (right column) at $\Lambda = 1.0\,\mathrm{nm}^{-1}$ and $k_BT = 2.0 \times 10^{-4}\,\mathrm{eV}$. The density assignments are (a, b) $\rho = +0.020$, (c, d) $\rho = +0.010$, (e, f) $\rho = -0.010$, (g, h) $\rho = -0.020$, (i, j) $\rho = -0.140$, and (k, l) $\rho = -0.150$, all in units of nm^{-2}. Black dashed vertical lines mark the LL-counting reference $F = 2\pi\rho_{\mathcal{S}}$, with $\rho_{\mathcal{S}} = |\rho|$ for the electron and shallow-hole sectors and $\rho_{\mathcal{S}} = \rho_{\mathrm{low},e}$ for the deep-hole sectors. The green dashed lines in panels (a) and (b) mark $F = \pi\rho$ for $\rho = +0.020\,\mathrm{nm}^{-2}$.

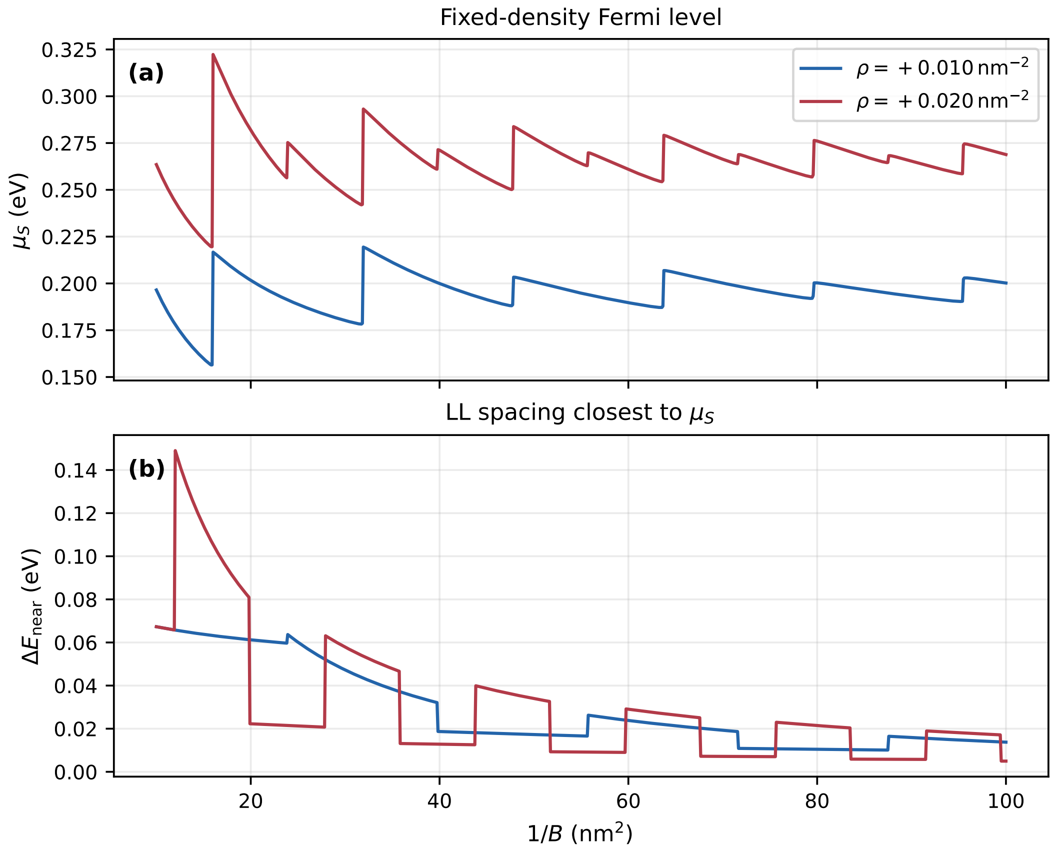

FIG. D.3. Comparison of the two electron-density cases used in Figs. D.1 and D.2. Panel (a) shows the fixed-density chemical potential $\mu_{\mathcal{S}}(1/B)$ for $\rho = +0.010$ and $+0.020\,\mathrm{nm}^{-2}$ at $k_BT = 2.0 \times 10^{-4}\,\mathrm{eV}$. Panel (b) shows the nearest-chemical-potential LL spacing $\Delta E_{\mathrm{near}} = E_{\mathrm{above}} - E_{\mathrm{below}}$, where $E_{\mathrm{below}} < \mu_{\mathcal{S}} < E_{\mathrm{above}}$ are the two LLs closest to the fixed-density chemical potential.

D.4. Local window selection

To resolve the thermal damping factor, we need to extract the oscillation amplitude as a function of temperature for the local $1/B$ windows of strong oscillation period, rather than a single global FFT. The windows are chosen from the lowest-temperature $\widetilde{M}_{\mathrm{no0LL}}$ trace and then held fixed for all temperatures. Operationally, we first identify zero crossings of the oscillatory trace and use same-slope zero crossings to define full local cycles. Fig. D.4 shows the resulting W_0, W_1, and W_2 windows. For $\rho = +0.020\,\mathrm{nm}^{-2}$, Fig. D.2(b) shows dominant magnetization weight near the lower frequency $F = \pi\rho$. The windows in Fig. D.4(a) therefore follow the corresponding dominant zero-to-zero cycles; the smaller shoulders are not treated as independent windows.

Table I compares the selected windows with the expected period $1/F$. For $\rho = +0.020\,\mathrm{nm}^{-2}$ we track only the main spin-degenerate-like LL oscillation in Fig. D.4(a), so the table uses $F = \pi\rho$. For the other densities, the windows follow the spin-resolved active-density counting frequency, $F = 2\pi\rho_{\mathcal{S}}$, with $\rho_{\mathcal{S}} = |\rho|$ for the electron and shallow-hole rows and $\rho_{\mathcal{S}} = \rho_{\mathrm{low},e}$ for the deep-hole rows. Because the numerical waveform is not a perfect sinusoid, individual zero-aligned windows can be slightly uneven; the final column checks the total span of the three selected windows against $3/F$ and gives a value close to unity for all densities, suggesting the correct assignment of the local windows to the expected oscillation period.

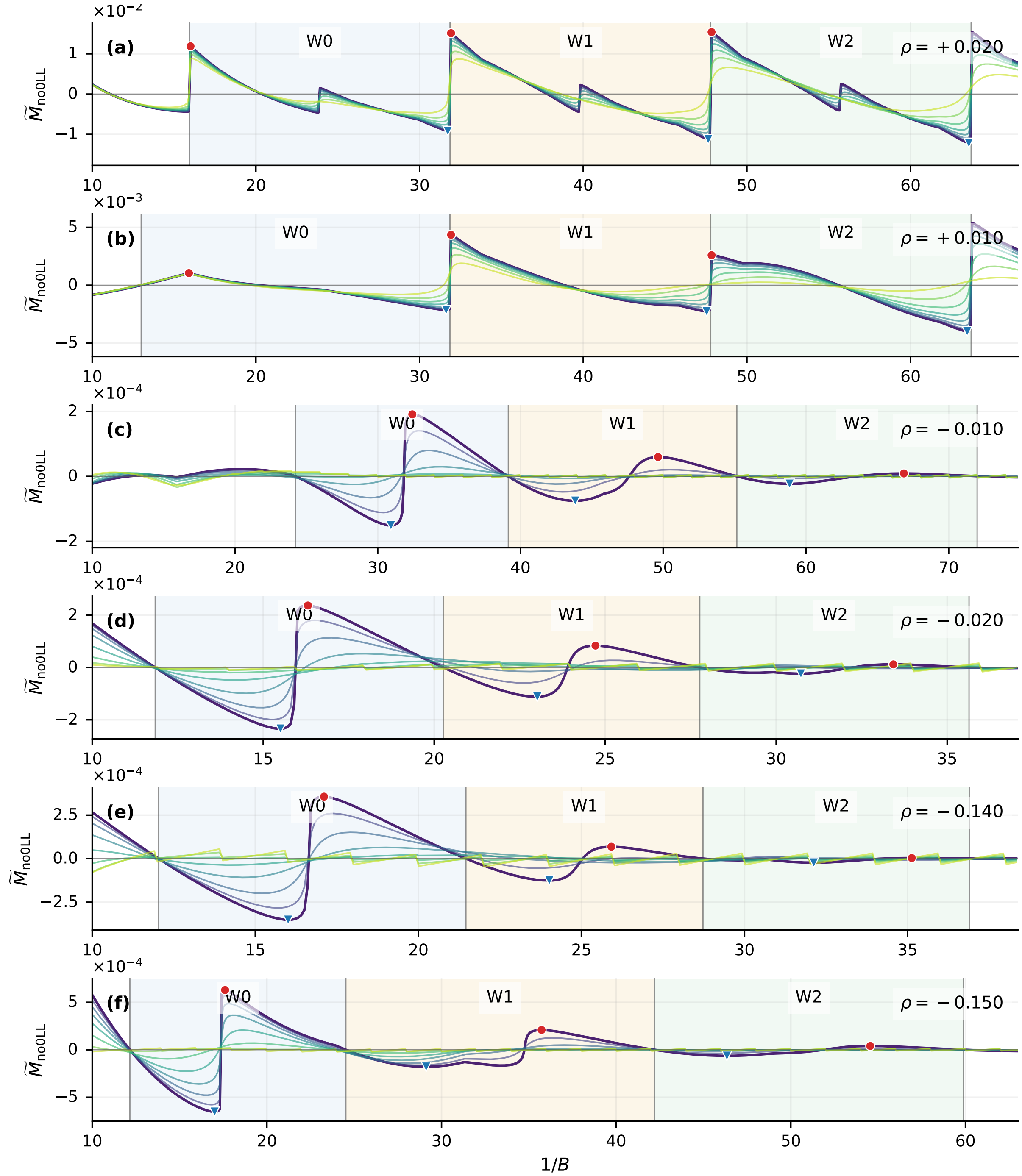

FIG. D.4. Local window selection for the zero-mode-excluded oscillatory magnetization at $\Lambda = 1.0\,\mathrm{nm}^{-1}$. Each row shows $\widetilde{M}_{\mathrm{no0LL}}(1/B)$ for one density, with the same temperature sequence as Fig. D.1. Shaded regions mark the three windows W_0, W_1, and W_2 used for local amplitude extraction; red circles and blue triangles mark the maximum and minimum of the lowest-temperature trace in each window.

TABLE I. Local window ranges in $x = 1/B$ for Fig. D.4. For $\rho = +0.020\,\mathrm{nm}^{-2}$, the period is evaluated using the main spin-degenerate-like frequency $F = \pi\rho$; for the other rows, it is evaluated using the spin-resolved active-density frequency $F = 2\pi\rho_{\mathcal{S}}$. The last column compares the combined three-window span with three ideal periods.

ρ	W_0	W_1	W_2	$1/F$	$\sum_i \Delta x_i/(3/F)$
$+0.020$	$[15.933, 31.862]$	$[31.862, 47.784]$	$[47.784, 63.703]$	15.915	1.000
$+0.010$	$[12.992, 31.857]$	$[31.857, 47.787]$	$[47.787, 63.700]$	15.915	1.062
-0.010	$[24.238, 39.150]$	$[39.150, 55.161]$	$[55.161, 71.995]$	15.915	1.000
-0.020	$[11.842, 20.265]$	$[20.265, 27.766]$	$[27.766, 35.639]$	7.958	0.997
-0.140	$[12.036, 21.459]$	$[21.459, 28.727]$	$[28.727, 36.892]$	8.309	0.997
-0.150	$[12.157, 24.523]$	$[24.523, 42.184]$	$[42.184, 59.880]$	17.385	0.915

D.5. Thermal damping of Normalized oscillation amplitude

Figure D.5 is obtained directly from the windowed oscillatory magnetization in Fig. D.4. For each density, temperature, and local window $w = W_0, W_1, W_2$, we first restrict the background-subtracted trace $\widetilde{M}_{\mathrm{no0LL}}(x,T)$, with $x = 1/B$, to the window interval $[x_{w,0}, x_{w,1}]$ listed in Table I. The frequency F_w is the same active-density counting frequency used to choose the window: for $\rho = +0.020\,\mathrm{nm}^{-2}$ it is the main spin-degenerate-like frequency $F_w = \pi\rho$, while for the other rows it is $F_w = 2\pi\rho_{\mathcal{S}}$. Within this fixed window, the fundamental $p = 1$ harmonic is extracted by the linear projection

$$\widetilde{M}_{\mathrm{no0LL}}(x,T) = C_{1,w}(T)\cos(2\pi F_w x) + S_{1,w}(T)\sin(2\pi F_w x) + C_{0,w}(T). \tag{D.13}$$

The numerical amplitude plotted by the solid markers is

$$\mathcal{R}^{\mathrm{num}}_{1,w}(T) = \frac{A_{1,w}(T)}{A_{1,w}(T_{\min})}, \qquad A_{1,w}(T) = \sqrt{C^2_{1,w}(T) + S^2_{1,w}(T)}. \tag{D.14}$$

For $\rho = -0.020$ and $-0.140\,\mathrm{nm}^{-2}$, the W_2 amplitudes are too small for a reliable normalized damping curve, so those two W_2 series are not shown.

The dashed curves in Fig. D.5 are then obtained by fitting the normalized amplitudes in each window to the standard LK thermal form

$$\mathcal{R}^{\mathrm{LK}}_{1,w}(T; m_w) = \frac{R_T(T, m_w, \bar{B}_w)}{R_T(T_{\min}, m_w, \bar{B}_w)}, \qquad R_T = \frac{X}{\sinh X}, \qquad X = \frac{2\pi^2 k_B T\, m_w}{\bar{B}_w}. \tag{D.15}$$

Here $\bar{B}_w = 1/[(x_{w,0} + x_{w,1})/2]$ is the representative field of the window, and m_w is the effective mass. Here we treat m_w as a single free fitting parameter, rather than using the branch-resolved analytical mass from Sec. C. The fitted m_w is summarized and compared with the branch-resolved expectation in Fig. D.6.

The branch-resolved LK expressions explain the scales and behaviors of these fitted masses. Using the local spacings from Sec. C, the LK thermal factor for a single branch is

$$R^{\alpha,\eta}_T(p,T) = \frac{X^{\alpha,\eta}_p}{\sinh X^{\alpha,\eta}_p}, \qquad X^{\alpha,\eta}_p = \frac{2\pi^2 p k_B T}{v^{\alpha,\eta}_\mu}. \tag{D.16}$$

Writing $\delta\mu = \mu - E_0$ and substituting the upper-branch spacings from Eq. (C.27) together with the lower-branch spacings from Eqs. (C.16) and (C.23), the corresponding branch-resolved arguments are

$$X^{\uparrow,+}_p = \frac{\pi^2 p k_B T}{aB}\frac{\delta\mu^2 - c^2 B}{\delta\mu^2}, \qquad X^{\downarrow,+}_p = \frac{\pi^2 p k_B T}{aB}\frac{\delta\mu^2 + c^2 B}{\delta\mu^2}, \tag{D.17}$$

$$X^{\uparrow,-}_p = \frac{\pi^2 p k_B T}{aB}\frac{c^2 B - \delta\mu^2}{\delta\mu^2}, \qquad X^{\downarrow,-}_p = \frac{\pi^2 p k_B T}{aB}\frac{\delta\mu^2 + c^2 B}{\delta\mu^2}. \tag{D.18}$$

The upper-branch expressions apply in the normal dispersive regime, while the lower-branch expressions apply in the anomalous regime within their corresponding crossing windows. For $\delta\mu^2 \gg c^2 B$, both normal expressions reduce to the parabolic result $X_p \simeq \pi^2 p k_B T/(aB)$. For anomalous LLs in the lower-branch crossing windows, $0 < \delta\mu < c\sqrt{B}$ for $E_-^{\uparrow}$ and $-c\sqrt{B} < \delta\mu < 0$ for $E_-^{\downarrow}$, the small local spacing makes X_p much larger at the same B and T. In the anomalous regime, where one lower branch dominates a given crossing window, the normalized damping is therefore well represented by the single-branch ratio

$$\mathcal{R}_{p=1}^{\alpha,-}(\rho, B, T) = \frac{R_T^{\alpha,-}(p=1, \rho, B, T)}{R_T^{\alpha,-}(p=1, \rho, B, T_{\min})}. \tag{D.19}$$

For normal electron doping, the two upper branches lie in the same energy range and are counted together at fixed density. The fixed-density chemical potential is obtained from

$$\rho = \frac{B}{2\pi} \sum_{\alpha=\uparrow,\downarrow} \sum_n f\big(E_+^{\alpha}(n, B) - \mu_\rho\big)\,, \tag{D.20}$$

and the two upper-branch contributions combine at the harmonic-amplitude level as

$$A_p^+(\rho, B, T) \propto \sum_{\alpha=\uparrow,\downarrow} v_\mu^{\alpha,+} R_T^{\alpha,+}(p, T) \cos\big(2\pi p n_{\alpha,+}^*\big)\,, \qquad \mathcal{R}_p^+(\rho, B, T) = \frac{\big|A_p^+(\rho, B, T)\big|}{\big|A_p^+(\rho, B, T_{\min})\big|}. \tag{D.21}$$

Here $E_+^{\alpha}(n_{\alpha,+}^*, B) = \mu_\rho$. Equation (D.21) gives the coherent physical harmonic expected from the analytical LK expression, whereas Fig. D.5 uses the simpler fitted form in Eq. (D.15) as a local numerical diagnostic of the thermal scale. The normal electron densities show weak thermal damping over this temperature range, whereas the anomalous-hole densities lose their $p = 1$ amplitude much more rapidly.

D.6. Effective masses from the windowed damping

The fitted standard-LK masses from the dashed curves in Fig. D.5 are summarized in Table II and Fig. D.6. The standard-LK fits in Fig. D.5 use only the local $p = 1$ harmonic and should therefore be read as qualitative thermal-scale diagnostics rather than precision measurements of a unique band mass. Because the numerical waveform is not a pure sinusoid, as clearly shown in Fig. D.4, the higher harmonics $p > 1$ is also not negligible, but we only focus on $p = 1$ harmonic in the current analysis of effective mass below.

Each marker in Fig. D.6 corresponds to the extracted effective mass for one window and is placed at the representative field $\bar{B}_w$. The curves show the branch-resolved analytical expectation $m_{\text{eff}}^*(B) = B/v_\mu$ from Sec. C; for normal electron densities we show both nearby upper branches, while for anomalous-hole densities we show the active lower branch selected by the fixed-density counting. Figure D.6(a) shows the effective mass for the normal LLs in electron carrier regime. Except for the first $\rho = +0.020\,\text{nm}^{-2}$ window, the fitted masses remain of order unity and lie around the parabolic scale $1/(2a)$, with visible window-to-window variation because the two upper branches, $E_+^{\uparrow}$ and $E_+^{\downarrow}$, contribute in the same energy range and are not exactly degenerate. For $\rho = +0.020\,\text{nm}^{-2}$, we note that the projected $p = 1$ amplitude is nearly temperature independent and even increases slightly with temperature, as shown in Fig. D.5a, so the one-parameter LK fit returns a mass close to zero. This suggests the higher harmonics ($p > 1$) also plays important role in the temperature dependence of the oscillation amplitude in this window, beyond the current $p = 1$ harmonics fitting. Fig. D.6(b) shows the effective mass of anomalous LLs in hole carrier regime. The fitted masses are much larger and more field dependent, qualitatively following the trend of the active lower-branch analytical curves. This is the mass-scale version of the rapid thermal damping in Fig. D.5: the anomalous LL spacing is much smaller, so the LK argument $X \propto m_{\text{eff}}^* T/B$ is much larger at comparable B and T. However, the fitted masses are quantitatively deviated from the analytical branch-resolved curves in Fig. D.6(b), suggesting the limitation of the effective mass description of the thermal damping in the anomalous LL regime.

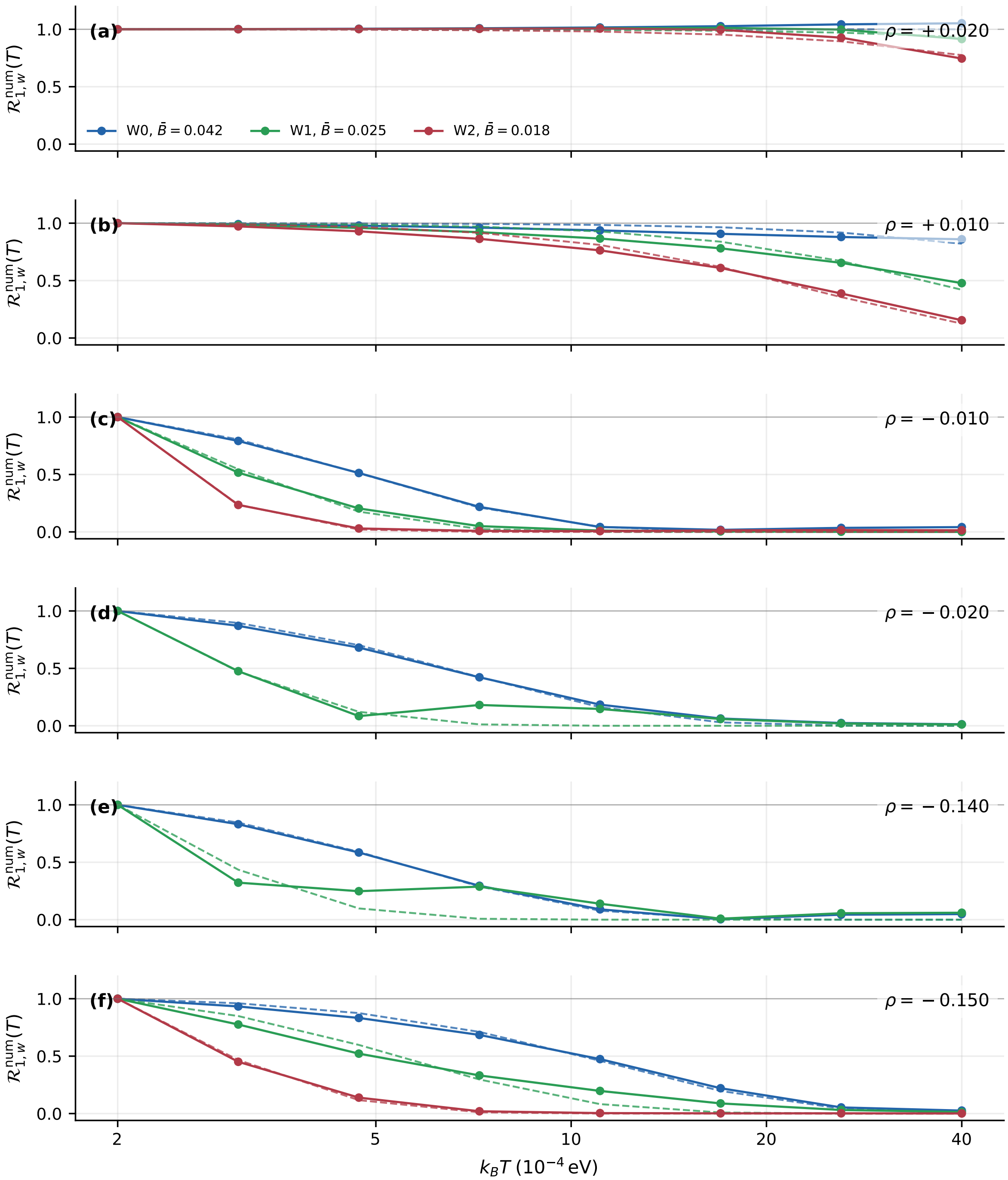

FIG. D.5. Windowed $p = 1$ thermal damping extracted from the Fig. D.4 windows. Solid markers show the normalized numerical amplitude $\mathcal{R}^{\mathrm{num}}_{1,w}(T)$ defined in Eq. (D.14); dashed curves are one-parameter standard-LK fits for the same window. Three colors blue, green and red label W_0, W_1, and W_2, respectively, and the legend gives the representative field $\bar{B}_w$ for the first row. The W_2 series for $\rho = -0.020$ and $-0.140\,\mathrm{nm}^{-2}$ are omitted because their amplitudes are too small for reliable damping extraction.

TABLE II. Fitted standard-LK masses m_w extracted from the dashed curves in Fig. D.5 and plotted in Fig. D.6. The entries correspond to the local windows W_0, W_1, W_2 defined in Fig. D.4.

$\rho\,(\mathrm{nm}^{-2})$	regime	m_{W_0}	m_{W_1}	m_{W_2}
+0.020	normal LL	1.04×10^{-4}	0.206	0.288
+0.010	normal LL	0.626	0.788	0.958
−0.010	anomalous LL	8.27	10.2	13.9
−0.020	anomalous LL	11.3	23.0	–
−0.140	anomalous LL	13.6	23.7	–
−0.150	anomalous LL	5.92	6.73	11.0

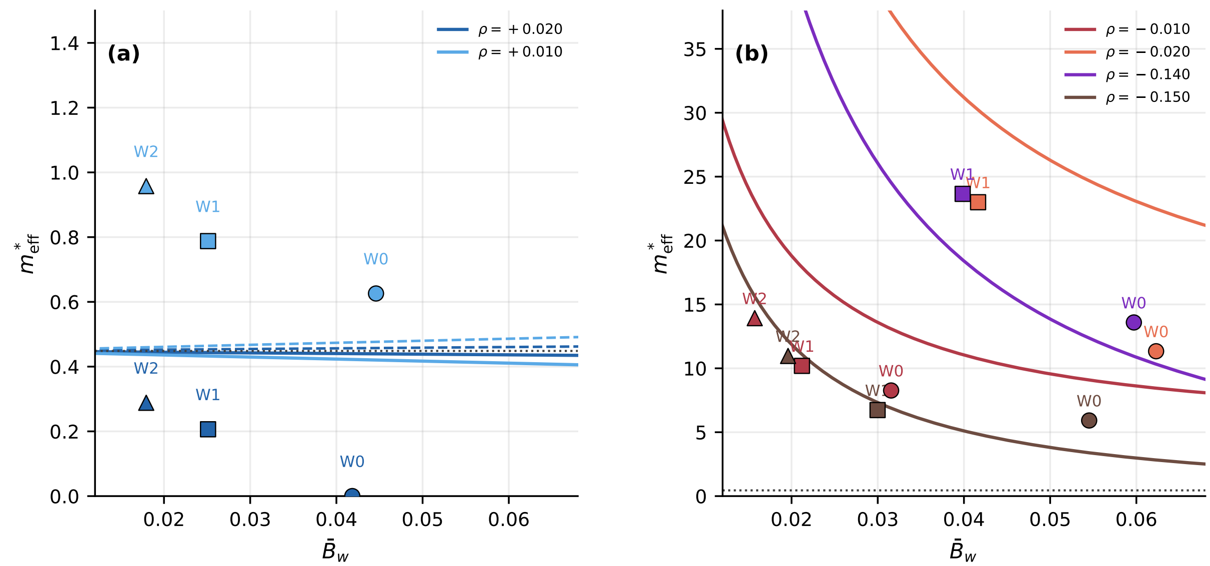

FIG. D.6. Effective masses extracted from the Fig. D.5 standard-LK fits. Panel (a) shows the normal electron windows for $\rho = +0.020$ and $+0.010\,\mathrm{nm}^{-2}$, with analytical curves for the nearby $E_+^{\uparrow}$ and $E_+^{\downarrow}$ upper branches; the near-zero W_0 point for $\rho = +0.020\,\mathrm{nm}^{-2}$ reflects the nearly non-damping projected $p = 1$ amplitude in that window. Panel (b) shows the anomalous-hole windows for $\rho = -0.010, -0.020, -0.140,$ and $-0.150\,\mathrm{nm}^{-2}$, with analytical curves for the active lower branch selected by the fixed-density counting; the W_2 points for $\rho = -0.020$ and $-0.140\,\mathrm{nm}^{-2}$ are omitted because their amplitudes are too small for reliable extraction. Markers are fitted masses for the local windows, labelled by W_0, W_1, W_2 when shown. The dotted horizontal line marks the parabolic value $1/(2a)$.